Ruolin Zhang

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EDUCATION

MFA in Drama and Film Studies, Documentary Track Sep 2022 – Jun 2025
Beijing Normal University, Beijing, China GPA: 3.8/4.0 (Awarded First-Class Scholarship)

BA in Chinese Language and Literature Sep 2018 – Jun 2022
North China University of Technology, Beijing, China GPA: 3.7/4.0 (Awarded as Outstanding Graduate)

RESEARCH PAPERS

Ruolin Zhang, Yue Li, Pan Hui. "Visiting Alone, but Not Alone: Exploring AI Guides and Peers in Solo VR Museum Visits." *(Under Review at ACM CHI 2027)*

Ruolin Zhang, Zheng Wei, Pan Hui. "The Convergence of Minds and Bodies: A Systematic Review of Generative AI Agents in Extended Reality for Education." *(Under Review at TVCG)*

Zheng Wei, Xiangze Meng, Jiayi Zhou, Zeqian Zhang, **Ruolin Zhang**, Juho Kim, Anyi Rao, Yun Wang. "RepoDirector: Decision-Carrying Storyboards for Repository-Grounded Generative Video Authoring." *(Under Review at ACM CHI 2027)*

Zhe Yan, Jingchen Gao, Zheng Wei, **Ruolin Zhang**, Junxiang Liao, Kehan Xie, Wai Tong, Lik-Hang Lee, Pan Hui. "The God-View Penalty: The Perspective Trade-Off in VR Cinematography Under Environmental Load." *(Manuscript in Preparation)*

RESEARCH EXPERIENCE

Research Assistant
Hong Kong University of Science and Technology (Guangzhou) | Advisor: Prof. Pan Hui

Generative AI Agents in XR Education & VR Cinematography Dec 2025 – May 2026

- Reviewed 73 studies from 2,029 records using PRISMA, developing a functional taxonomy of GenAI agent roles and design implications for XR education.
- Conducted a 2×2 VR user study examining the effects of viewing perspective and environmental load using NASA-TLX and stimulated-recall interviews.

AI Agents in Solo VR Museum Visits May 2026 – Sep 2026

- Designed and implemented an LLM-powered VR museum comparing five agent configurations: No Agent, Peer, Guide, Integrated, and Dual Agents.
- Conducted a within-subjects mixed-methods study with 27 participants, evaluating companionship, knowledge gain, and interaction burden through questionnaires, knowledge tests, behavioral data, and interviews.

Master's Researcher Jan 2025 – Jun 2025
Beijing Normal University | Advisor: Prof. Qipeng Fan
Master's Thesis: Interface as Performance: Application of Desktop Elements in the Documentary

- Analyzed how UI elements and screen-based interactions structure narratives, guide attention, and manage cognitive load in desktop documentaries.
- Theorized interface navigation as a "performative act" that shapes user perception, trust, and digitally mediated social contexts.

SKILLS

Technical Skills: Python, Unity Development, Multi-agent Framework Development
Design Skills: Figma, Procreate, Blender, Interaction & User Flow Design
Research Methods: Mixed-Methods Evaluation, Experimental Design, Semi-structured Interviews, Thematic Analysis, Telemetry Data Analysis, NASA-TLX